BT-204 (GS)

B.Tech., I & II Semester Examination, June 2026

Grading System (GS)

BASIC CIVIL ENGINEERING & MECHANICS

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Full paper (Q1–Q8)

1. (a) Explain the properties, laboratory tests and engineering applications of cement. Discuss how these properties influence the quality of concrete. **7**

(b) Describe the types, properties and engineering uses of bricks and stones used in building construction. **7**

2. (a) Explain the construction and suitability of spread footing and RCC footing with neat sketches. **7**

(b) Describe the principle, construction and applications of theodolite and electronic surveying instruments in modern surveying. **7**

3. (a) Explain the methods of measurement of distances and elevations in surveying. Compare conventional methods with EDM. **7**

(b) During levelling, the following staff readings were observed: B.S. = 1.425 m and F.S. = 2.165 m. If the Reduced Level (R.L.) of the benchmark is 100.000 m, determine the Reduced Level of the next point by the Height of Instrument method. **7**

4. (a) Explain the principle of contouring and discuss its applications in civil engineering projects. **7**

(b) Describe the applications of remote sensing in civil engineering and explain its advantages over conventional surveying. **7**

5. (a) Explain the methods of area and volume computation in surveying. Discuss their applications in earthwork estimation. **7**

(b) Explain the conditions of equilibrium of coplanar forces. Discuss the significance of Free Body Diagram (FBD) in solving engineering mechanics problems. **7**

6. (a) Explain the Method of Joints for the analysis of plane trusses. State its assumptions and limitations. **7**

(b) Explain the laws of dry friction and discuss their engineering applications with suitable examples. **7**

7. (a) Explain the concept of Centroid and Centre of Gravity. Differentiate between them with suitable examples. **7**

(b) Explain the procedure for drawing the Shear Force Diagram (SFD) and Bending Moment Diagram (BMD) for a simply supported beam subjected to concentrated and uniformly distributed loads. **7**

8. Write a short note on any two of the following: **14**

(a) Radius of Gyration and its engineering significance (b) Plastering and pointing

(c) Reciprocal Levelling (d) Survey Stations and their classification

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